

Introduction

Rock, paper, scissors is a famous game used to settle disputes, because it is seemingly fair odds. It models a predator-prey relationship between entities, where scissors is crushed by rock, cuts paper, and paper folds over/surrounds rock. My agent-based model simulates a game of rock, paper, scissors—except there are adjustable settings which imbalance the teams.

These settings: starting populations per agent team/turtle breed and a vision radius for scissors are the center of two experimental research questions:

1. How do different starting populations for scissors affect the outcome of which agent comes out on top?
2. How do different vision radius ranges affect the outcome of which agent comes out on top?

Methodology

The model consists of three agent types—rock, paper, and scissors—randomly distributed across a wrapped 2D grid environment. Each agent moves forward and interacts with others upon collision. When two agents meet, the losing type is converted into the winning type (e.g., scissors becomes rock when encountering rock).

Experiment 1: Starting Population

- Rock and paper populations fixed at 20 each
- Scissors population varied from 10 to 50 (increments of 10)
- Scissors vision disabled
- 10 simulations per condition

Experiment 2: Vision Radius

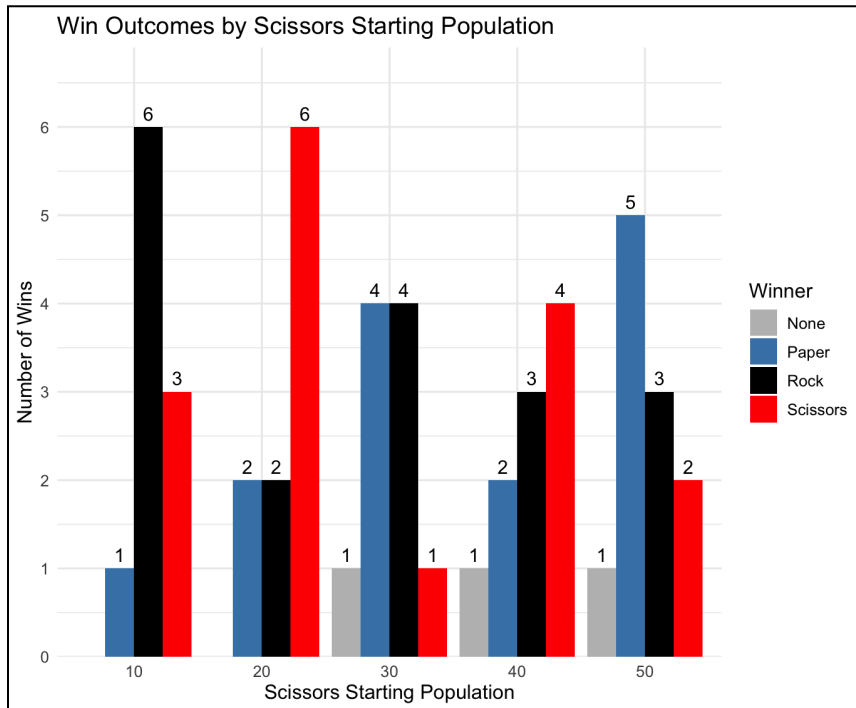
- All populations set to 30
- Scissors vision radius varied from 0 to 30 (increments of 10)
- Vision allows scissors to chase prey and avoid predators
- 10 simulations per condition

Each simulation ran until one agent type dominated or until 1500 ticks were reached.

I collected my data by running simulations through Behavior Space on NetLogo then manually tallying win counts per team/non-team. Then I created dataframes to visualize this data in R.

Results

Experiment 1:

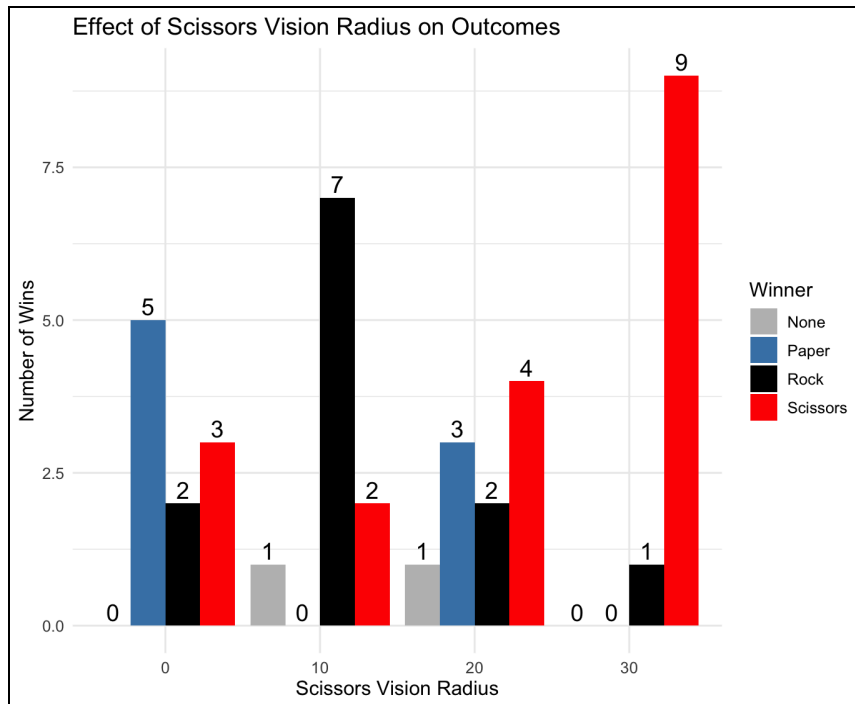


In the first set of 10 simulations, scissors is outnumbered by 10 turtle agents compared to both rock and paper. Rock wins 6/10 times.

In the second set of 10 simulations, the scissors population is set at 20, that is when the match should be equal and fair, since the other teams, rock and paper, also start with twenty. Coincidental or strange, this is when scissors dominated most wins out of 10 simulations.

In the next three sets of simulations, scissors increased outnumbers rock and paper by additions of 10 turtles during set-up. It does poorly starting with 30, better at 40 (but almost relatively equal to its competition), and loses majority again at 50.

Experiment 2:



At the baseline of zero vision radius for team scissors, paper wins 5/10 times. This simulation had no strategy, equal circumstances, so no consistent winner. I believe if this simulation were recorded many times, more than 10, an even competition would show.

With a vision radius of ten, scissors intentionally eat up the paper population. But with the variation of random 1-direction, forward movements of rocks (and papers), rock agents are spread out, covering the map in most areas. Therefore, it is difficult for scissors to flee/avoid them with vision radiuses on 10. Some get caught, escaping one rock's path but unintentionally entering another's. Eventually, with no paper to check rock, scissors are wiped out.

With vision radiuses of 20 and above, especially as observed in 30, scissors become a lot better at avoiding rocking and targeting paper. Their movements are more intentional and short, less consistent forward steps and more constant turning and redirecting. When paper and rock agents are close to each other, they avoid, since both are in their radiuses. This allows for when many papers have been depleted, the game just becomes paper vs rock whenever they coincidentally collide (since their settings are defaulted to moving forward always) with scissors observing on the sidelines. Eventually paper will take over rock as its natural predator and fall prey to the mob of scissors.

Conclusion

Experiment 1:

Increasing the starting population of scissors did not result in scissors dominance due to the cyclical predator-prey nature of the model. While higher numbers of scissors initially reduced the

paper population, this indirectly benefited rock, which preys on scissors. As a result, larger scissors populations created more opportunities for rock to expand. It becomes counter-productive and self-harm for the scissors team. This experiment demonstrates that in cyclic competitive systems, increasing one agent type can unintentionally strengthen its predator rather than itself.

Experiment 2:

Upping the vision radius of scissors is increasingly advantageous—*very advantageous*. As vision increases, scissors are able to more effectively target paper while avoiding rock, transforming from a random agent into a strategic one. At higher levels, this behavioral advantage disrupts the natural cyclical predator-prey balance of the model and leads to consistent scissors dominance. While I did not record this for experiments, even when outnumbering scissors, scissors are able to most often come out on top, dominating competition, when the vision radius is high. That was my observation for an unofficial sensitivity analysis.

References

- Wilensky, U. (1999). *NetLogo* [Computer software]. Center for Connected Learning and Computer-Based Modeling, Northwestern University. <http://ccl.northwestern.edu/netlogo/>
- Wilensky, U. (1997). *NetLogo Wolf Sheep Predation model* [Computer software]. Center for Connected Learning and Computer-Based Modeling, Northwestern University. <http://ccl.northwestern.edu/netlogo/models/WolfSheepPredation>